

### Project Initialization and Planning Phase

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 28 June 2026                                                |
| Team ID       | SWTID-2026-4836                                             |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 3 Marks                                                     |

**Define Problem Statements:**

Farmers often face difficulties in selecting the most suitable crop due to a lack of knowledge about soil nutrients and changing environmental conditions. Incorrect crop selection leads to low productivity, resource wastage, and financial losses. Therefore, there is a need for an intelligent machine learning-based system that analyzes soil and environmental parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop and support sustainable agricultural decision-making.

| Problem Statement (PS) | I am (Customer)                                 | I'm trying to                                                   | But                                                                           | Because                                                                        | Which makes me feel                                    |
|------------------------|-------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------|
| PS-1                   | A farmer seeking suitable crops for cultivation | Select the best crop based on soil and environmental conditions | I don't know which crop is suitable for my land and weather conditions        | I lack proper knowledge about soil nutrients, rainfall, and climate conditions | Worried about crop failure and financial losses        |
| PS-2                   | An agricultural researcher                      | Analyze agricultural data to improve farming decisions          | Processing large amounts of soil and environmental data manually is difficult | Traditional methods are time-consuming and less accurate                       | Frustrated and limited in making data-driven decisions |
| PS-3                   | A small-scale farmer                            | Increase crop productivity and profit                           | I often choose crops based on guesswork                                       | I do not have access to intelligent farming guidance                           | Uncertain about my farming future                      |